CASE STUDY / GREENFIELD PLANT / DEEPAK CHEM TECH

# A connected foundation for a new plant.

## Deepak Chem Tech Limited

**Dahej-3 | Industrial IT infrastructure**

Engineered connectivity through fibre, structured cabling, centrally managed Aruba Wi-Fi and Cambium wireless links for distributed plant points.

**PROJECT VALUE: APPROX. USD 36,299**

PO 4344000056 | Supply, installation & engineering

Tax-inclusive PO equivalent; 18 March 2026 FX basis.
Design narrative based on the supplied order and site photograph.



# Scope & scale | A coordinated infrastructure package

The greenfield setup combines local wired connectivity, indoor wireless access, inter-area fibre and fixed wireless transmission. D-TECH’s delivery scope is presented from design and engineering through supply, installation, testing and handover. This document explains the ordered solution; completed installation and measured results require site acceptance records.

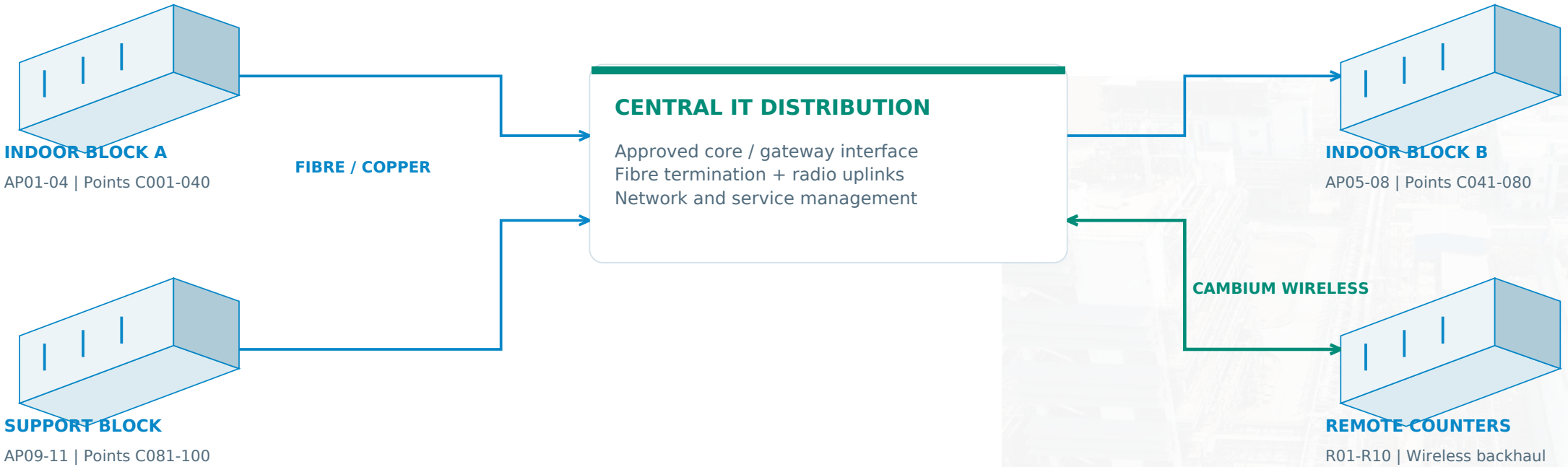
| Workstream | Ordered baseline | Design role |
|---|---|---|
| Wi-Fi & management | 11 AP-515 APs; 11 brackets; 12 30W midspans; 11 Aruba Central Foundation 5-year licences; 11 five-year AP support services. | Indoor Wi-Fi access, central provisioning and monitored AP operation. |
| Fixed wireless | 2 ePMP 4500L base radios; 2 sector antennas; 10 Force 425 subscriber radios; 12 surge suppressors; 2 cnMaestro X one-year licences. | Distributed Ethernet backhaul; confirm subscriber-management entitlement. |
| Fibre & field racks | 1,000 m 6-strand armoured OFC; 1,000 m HDPE; 24 LIUs; 24 LC pigtails; 5 x 9U + 10 x 6U racks. | Protected inter-area paths, termination and equipment locations. |
| Counter / desk network | 15 Cat6 boxes; 100 faceplates / back boxes; 200 RJ45 jacks; 100 x 1 m patch cords; 5 x DGS-108 switches. | User-side connectivity; outlet schedule and switch port plan to reconcile. |

03 Campus sketch | 04 End-to-end network | 05 Aruba Wi-Fi zones | 06 Wireless backhaul | 07 Material checks | 08 Delivery & support.

ILLUSTRATIVE 2D / 3D DRAWING / LOCATION LABELS FOR DESIGN EXPLANATION

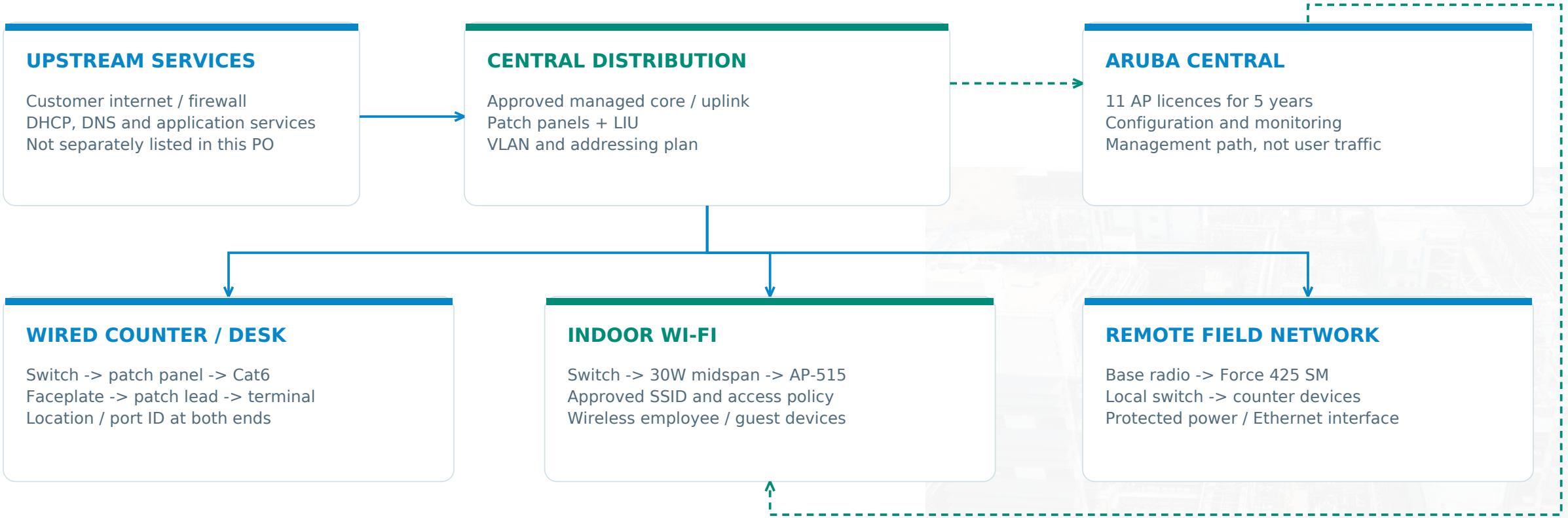
# Campus sketch | Connected counters and Wi-Fi zones

A central IT distribution point connects building networks and remote counters. Counter labels below group 100 planned faceplate points for readability; they do not establish actual rooms, departments or commissioned user quantities.




AP group allocations and building positions are conceptual. Survey determines actual indoor zones, counter mapping, line of sight and rack locations.

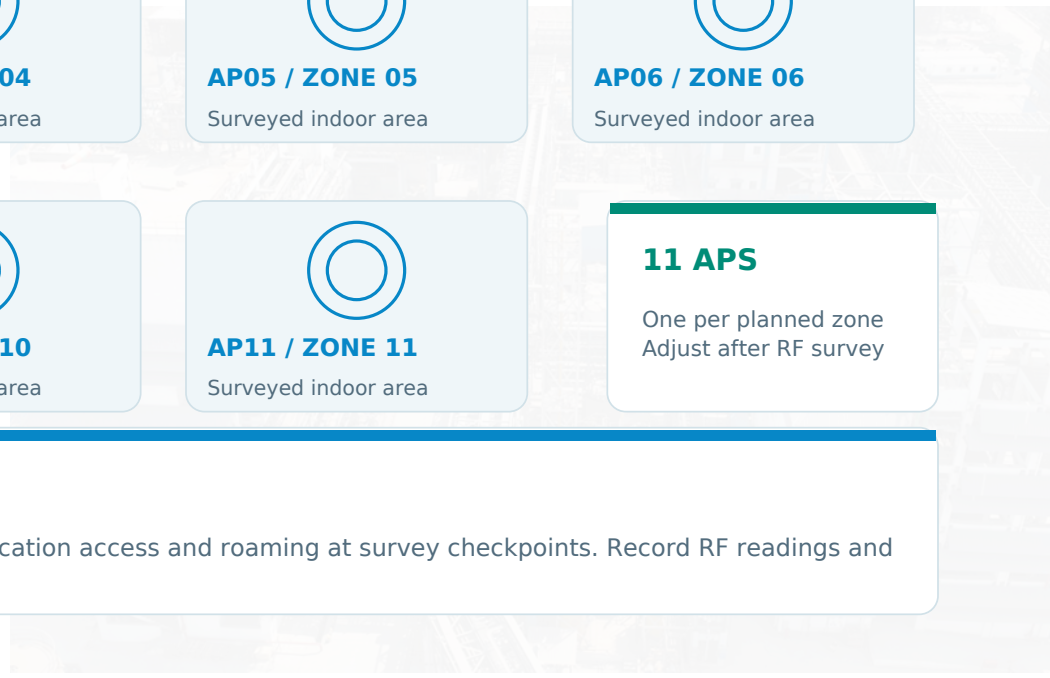
# Network schematic | Every point follows a traceable path




Core switching, firewall and servers are integration dependencies, not confirmed supply lines. DGS-108 units are unmanaged and cannot provide per-port VLAN policy.

# HPE Aruba Wi-Fi | Eleven indoor access zones

Plan AP locations against usable indoor space and client demand. The ordered AP-515 is an indoor campus access point; the layout below shows eleven logical zones, not measured RF coverage. Mounting and configuration follow the approved deployment mode and site plan.



**AP01 / ZONE 01**
Surveyed indoor area

**AP02 / ZONE 02**
Surveyed indoor area

**AP03 / ZONE 03**
Surveyed indoor area

**AP04 / ZONE 04**
Surveyed indoor area

**AP05 / ZONE 05**
Surveyed indoor area

**AP06 / ZONE 06**
Surveyed indoor area

**AP07 / ZONE 07**
Surveyed indoor area

**AP08 / ZONE 08**
Surveyed indoor area

**AP09 / ZONE 09**
Surveyed indoor area

**AP10 / ZONE 10**
Surveyed indoor area

**AP11 / ZONE 11**
Surveyed indoor area

**11 APS**

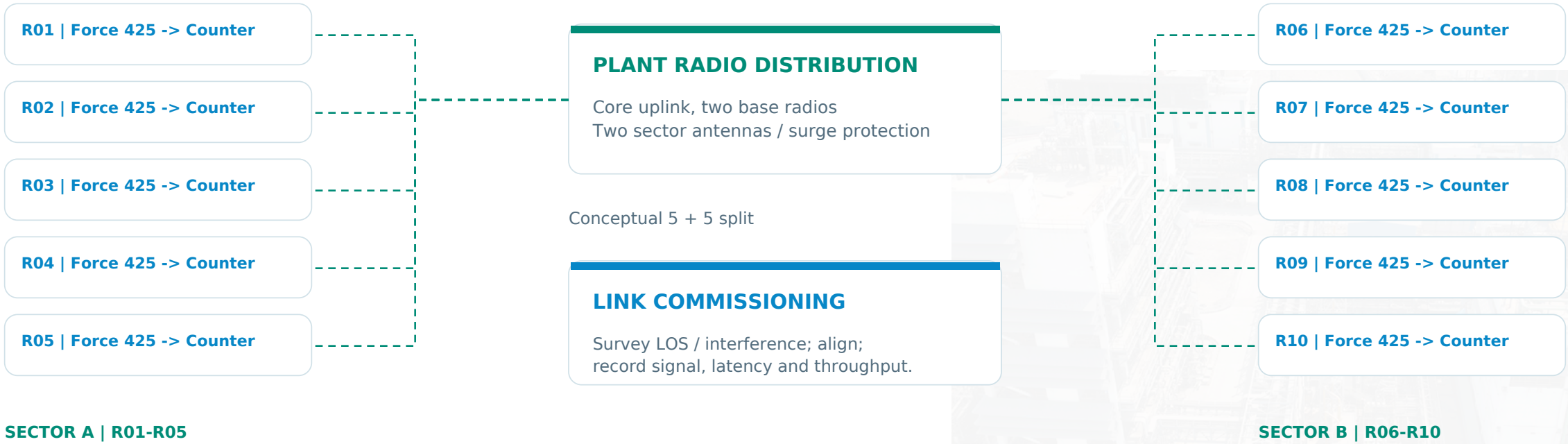
One per planned zone
Adjust after RF survey

## CONFIGURATION & ACCEPTANCE

Separate employee / guest access as approved; verify AP power, management reachability, authentication, application access and roaming at survey checkpoints. Record RF readings and final AP mapping.

# Wireless backhaul | Two sectors, ten remote nodes

The ordered mix supports a point-to-multipoint concept: two ePMP 4500L base radios serve ten Force 425 subscriber nodes. Each drawn path represents a radio link; a dedicated P2P arrangement needs separate radio-mode and endpoint allocation approval.




No radio distance or measured throughput is specified in the PO. Confirm path profile, power, mounting, surge bonding and licences for all managed nodes.

# Material review | Resolve interfaces before installation

| Review point | Finding in supplied order | Required resolution |
|---|---|---|
| Fibre optics & SFPs | 25 modules are explicitly 1G copper SFPs; fibre accessories include LC single-mode and multimode patch leads. | Confirm fibre type / link budgets and compatible optical transceivers; copper SFPs cannot terminate optical fibre. |
| AP bundles & licences | AP description mentions bracket / injector; 11 brackets and 12 midspans also appear as separate lines. | Reconcile bundled versus separate delivery; confirm spare injector allocation and licence registration / start dates. |
| Switching & ports | 5 unmanaged 8-port switches versus 100 faceplates, 200 jacks and multiple panels. | Produce port schedule; identify customer core / managed switches and any capacity gap. These quantities are not 200 users. |
| Cable / termination units | 15 Cat6 boxes and 4,575 m laying; separate armoured cable line says 5 m at INR 33,000/m with 1,525 m laying. | Verify box lengths and correct armoured supply UOM / quantity against route measurements; reconcile LIU / pigtail counts. |
| Civil & rack work | 100 generic rack-dressing units plus 5 / 10 rack-specific units; RCC piling specification accompanies a 10 m quantity. | Check duplicate activity billing and actual civil specification; obtain approved route and foundation drawings. |
| Radio management | 2 cnMaestro X AP licences for one year; 2 base radios and 10 SMs. | Verify applicable subscriber entitlement / platform model and renewal ownership; document all 12 protected endpoints. |

These are reconciliation actions, not confirmed shortages or defects. Obtain approved BOM and drawings before dispatch / configuration.

DESIGN / DEPLOYMENT / TESTING / CUSTOMER HANDOVER

# Delivery & operations | A controlled implementation path

| Stage | D-TECH delivery process / acceptance record |
|---|---|
| 01 \| Survey & approve | Map counters, racks, cable routes, Wi-Fi zones and radio paths; agree addressing, security policy and a reconciled BOM. |
| 02 \| Supply & install | Check model / serial / quantity; obtain dispatch clearance; install containment, fibre, racks, patching, APs and radio endpoints safely. |
| 03 \| Configure & test | Validate cabling / fibre reports, approved segmentation, AP licences and radio links. Test terminal reachability, Wi-Fi access and application paths. |
| 04 \| Hand over & support | Submit as-built drawings, port / AP / radio registers, configuration copies, test reports, warranty certificates and open-action ownership. |

**D-TECH Customer Care | support.dtechindia.com | +91 99989 03042**

Customer: Deepak Chem Tech Limited, Dahej-3, New Survey No. 142, Padariya, Vagara, Gujarat. PO 4344000056 dated 18 March 2026.

Value basis: tax-inclusive INR 3,383,979.70 / 93.22515 INR per USD (18 March 2026, Exchange-rates.org), rounded; presentation equivalent only.

References: supplied PO and site images; HPE Aruba 510 Series documentation; Cambium ePMP 4500L planning guide. Diagrams are conceptual.

Five-year Aruba services are ordered OEM entitlements, not a blanket D-TECH AMC. Verify activation, eligibility and applicable support terms.